Note that in the prior counter example, K had a "hole" in which f could hide a pole so that f would have a nonzero integral. So, a question one can ask is whether that hole is the only thing preventing us from being able to generally approximate functions in $O(K)$ with polynomials?

In this class, if $K \subseteq \mathbb{C}$ is a compact set then we define a hole of K to be a bounded connected component of $\mathbb{C} - K$.

One note is that as K is closed, $\mathbb{C} - K$ is open and thus locally connected. It follows that any hole H is an open region.

Also note that a compact set $K \subseteq \mathbb{C}$ has no holes iff $\mathbb{C} - K$ is connected. After all, K being compact implies that $\mathbb{C} - K$ has exactly one unbounded connected component. Then K is hole-free iff $\mathbb{C} - K$ equals that unique unbounded component.

Proposition: If $K \subseteq \mathbb{C}$ is compact then every $f \in O(K)$ can by uniformly approximated with polynomials only if K is hole-free.

Proof:
Suppose K has a hole H and fix $a \in H$. Then set $f(z) = \frac{1}{z-a}$ and let $M = \max_{z \in K} |z - a| > 0$. If $\{p_n\}_{n \in \mathbb{N}}$ is a sequence of polynomials uniformly converging to f, then we can find N such that $|p_N - \frac{1}{z-a}| < \frac{1}{2M}$ in K. In turn:

$$|(z-a)p_N - 1| < \frac{|z-a|}{2M} \leq \frac{1}{2} \text{ in } K.$$

Yet now $g(z) := 1 - (z-a)P_N$ is a polynomial on $\overline{H}$ satisfying that $g(a) = 1$ and $|g(z)| < \frac{1}{2}$ on $\partial H \subseteq K$. This contradicts the maximum modulus principle. So, we've proven that if K is not hole-free then there is always some function $f \in O(K)$ which cannot be uniformly approximated with polynomials. ■

As a corollary to the theorem we are about to prove though, we can also say that the converse of the prior proposition holds. Namely:

Runge's Little's Theorem: If $K \subseteq \mathbb{C}$ is compact and hole-free, then for all $f \in O(K)$ there exists polynomials $\{p_n\}_{n \in \mathbb{N}}$ with $p_n \xrightarrow{u.} f$ on K.

As an interesting corollary to Runge's little theorem, if A is any simple proper subarc of S^1 then we can find a polynomial $p(z)$ with $p(0) = 1$ and $|p(z)| < 1$ for all $z \in A$.

After all we can pick a polynomial $q(z)$ with $|q(z) - \frac{1}{z}| < 1$ for all $z \in A$. In turn $|1 - zq(z)| < |z| = 1$ for all $z \in A$. So, set $p(z) = 1 - zq(z)$. Then $p(0) = 1$ and $p(z) < 1$ for all $z \in A$.

Returning to motivating Runge's theorem, perhaps our issue preventing us from getting a far reaching general theorem is that we're trying to pick from too narrow of a set of functions. If an issue we've run into is that we need our approximations to have poles in the holes, then perhaps we should instead be trying to find rational function approximations.

Namely, if $K \subseteq \mathbb{C}$ is compact and $f \in O(K)$, then we should try to uniformly approximate f with a rational function whose poles are at prescribed points in the holes of K. In other words, we'd want to prove the following:

Theorem (almost final version): Suppose $K \subseteq \mathbb{C}$ is compact and let $S \subseteq \mathbb{C}$ be a set of points in $\mathbb{C}$ such that S contains at least one point from each hole of K. Then for any $f \in O(K)$ we can find a sequence of rational functions $\{R_n\}_{n\in\mathbb{N}}$ such that each R_n has poles only in S and $R_n \xrightarrow{u.} f$ on K.

Two Remarks:

- The R_n don't need to have poles at every point in S.
- If K has no poles then we can take $S = \emptyset$. Then none of the R_n have poles, meaning they are all polynomials. Thus, we recover Runge's little theorem.

To get the actual theorem statement that we will prove, we replace $\mathbb{C}$ with $\mathbb{C}_\infty$.

We say f has a pole at ∞ if $f(\frac{1}{z})$ has a pole at $z = 0$.

Note that any rational function $\frac{p(z)}{q(z)}$ has a pole at ∞ iff $\deg(p) > \deg(q)$. In particular, $\frac{p(z)}{q(z)}$ only has poles at ∞ if $\deg(q) = 0$ and $\deg(p) > 0$. In other words, if a rational function only has a pole at ∞, then it is a nonconstant polynomial. Similarly, we can see that a rational function with no poles at all (not even at ∞) must be a constant function.

Runge's Theorem (for compact sets): Suppose $K \subseteq \mathbb{C}$ is compact and $\widehat{S} \subseteq \mathbb{C}_\infty$ contains a single point from each connected component of $\mathbb{C}_\infty - K$. Then for all $f \in O(K)$ and $\varepsilon > 0$ there exists a a rational function R with poles only in $\widehat{S}$ such that $|f - R| < \varepsilon$ on K.

Hopefully it's clear that the almost final version of the theorem is recovered when we choose ∞ to be the element of $\widehat{S}$ in the unbounded (when restricted to $\mathbb{C}$) component of $\mathbb{C}_\infty - K$.

Proof:
Our general strategy will be to first establish an integral formula for compact sets. Secondly, we shall prove the existence of rational approximations without prescribed poles. Finally, we will push the poles to our desired locations.

Lemma 1: Let $K \subseteq U \subseteq \mathbb{C}$ where K is compact and U is an open set. Then there exists oriented line segments $\sigma_1, \ldots, \sigma_n$ in $U - K$ satisfying that for all holomorphic functions $f \in O(U)$ we have that $f(a) = \frac{1}{2\pi i}\sum_{j=1}^{n}\int_{\sigma_j}\frac{f(z)}{z-a}\mathrm{d}z$ for all $a \in K$.

Proof:
Since K is bounded, by restricting U we can without loss of generality assume that $U \neq \mathbb{C}$. This is important as it let's us then conclude that $\mathbb{C} - U$ is a nonempty closed set disjoint from K, and in turn $d := \inf\{|x - y| : x \in K, y \in \mathbb{C} - U\} > 0$.

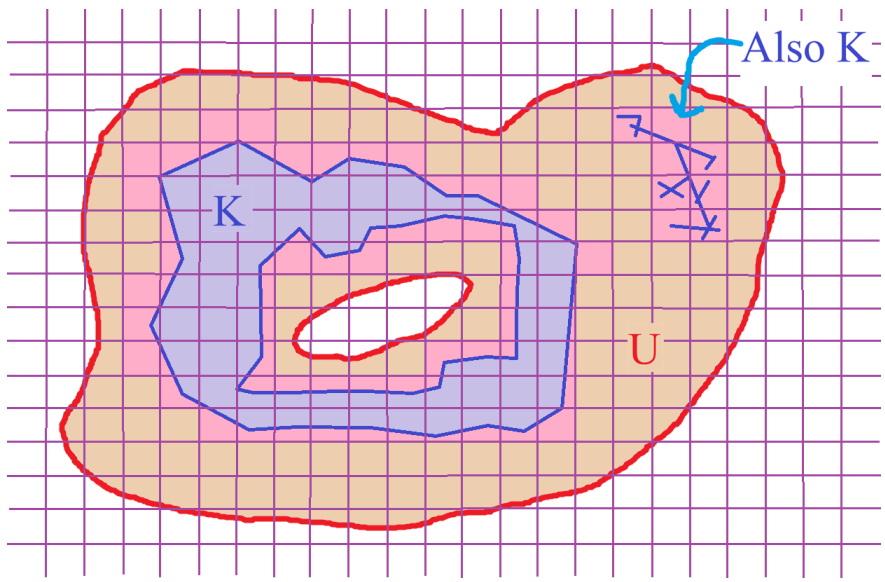
Tile $\mathbb{C}$ with a grid of squares of side length $\frac{d}{2\sqrt{2}}$. Then consider the closed squares $Q_1, Q_2, \ldots, Q_m$ that intersect K. Since K is compact and thus bounded, we know that there are only finitely many such squares.

Claim 1: $K \subseteq \bigcup_{j=1}^{m} Q_j \subseteq U$.

Every point in $\mathbb{C}$ is contained in at least one square of our grid. As a result, each $k \in K$ is contained in some square Q. And by how we picked the Q_j we then know that that Q is one of those Q_j. So, this gives us that $K \subseteq \bigcup_{j=1}^{m} Q_j$.

Next note that $\operatorname{diam}(Q_j) = {}^{d}/_{2}$ for each j. Thus because of how we defined d plus the fact that each Q_j contains an element of K, we know that $Q_j \subseteq U$ for all j.

Below is a picture that hopefully makes it clear what U, K our tiling of squares, and the Q_j we picked might look like.



We can orient the boundary of each square in our tiling to be traveling counter-clockwise. Then, let $\sigma_1, \ldots, \sigma_n$ be the edges of $\partial Q_1, \ldots, \partial Q_m$ which are not shared by distinct squares Q_{j_1}, Q_{j_2} where $1 \leq j_1 < j_2 \leq m$.

Claim 2: $\bigcup_{j=1}^{n} \{\sigma_j\} \subseteq U - K$.

Note that $\{\sigma_j\} \subseteq U$ by claim 1. Meanwhile, assume $\{\sigma_j\} \cap K \neq \emptyset$ and let $k \in \{\sigma_j\} \cap K$. Since each edge in our tiling is shared by two squares, we thus have that K intercepts both of those squares. And now we have a contradiction since $\{\sigma_j\}$ borders two distinct Q_j.

Now suppose $f \in O(U)$.

Claim 3: For all $a \in U - \bigcup_{j=1}^{m} \partial Q_j$, we have that:

$$\sum_{j=1}^{m} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{f(z)}{z-a} \mathrm{d}z = \sum_{j=1}^{n} \frac{1}{2\pi j} \int_{\sigma_j} \frac{f(z)}{z-a} \mathrm{d}z$$

This is obvious because the common sides of the Q_j's will be oriented in opposite directions and thus cancel out, leaving only the integrals over the σ_j.

Note that if $a \in Q_\ell^\circ$, then by Cauchy's integral formula (see *pages 402-404*):

$$\frac{1}{2\pi i}\int_{\partial Q_j}\frac{f(z)}{z-a}\mathrm{d}z = \begin{cases} f(a) & \text{if } j=\ell \\ 0 & \text{otherwise} \end{cases}$$

In turn by claim 3, we can conclude for all $a \in \bigcup_{j=1}^m Q_j^\circ$ that:

$$f(a) = \frac{1}{2\pi i}\sum_{j=1}^n \int_{\sigma_j}\frac{f(z)}{z-a}\mathrm{d}z$$

Finally, note that $g(a) := \frac{1}{2\pi i}\sum_{j=1}^n \int_{\sigma_j}\frac{f(z)}{z-a}\mathrm{d}z$ is a continuous function from the set $\mathbb{C} - \bigcup_{j=1}^n \{\sigma_j\}$ (see Leibniz's rule on *pages 380-381* for why g is continuous). Also g agrees with the continuous function f on a dense subset of $(\bigcup_{j=1}^m Q_j) - (\bigcup_{j=1}^n \{\sigma_j\})$ due to the above formula. So, $f = g$ everywhere on $(\bigcup_{j=1}^m Q_j) - (\bigcup_{j=1}^n \{\sigma_j\}) \supseteq K$. This proves our lemma.

Now given any fixed $f \in O(K)$ and $\varepsilon > 0$, note by definition that f can be holomorphically exteded to some open set U containing K. In turn, by the prior lemma we can pick line segments $\sigma_1, \ldots, \sigma_n$ in $U - k$ satisfying for all holomorphic functions $g \in O(U)$ that $g(a) = \frac{1}{2\pi i}\sum_{j=1}^n \int_{\sigma_j}\frac{g(z)}{z-a}\mathrm{d}z$ for all $a \in K$.

Define $F_j(z) := \frac{1}{2\pi i}\int_{\sigma_j}\frac{f(w)}{w-z}\mathrm{d}w$ for each j and note that each F_j is a holomorphic function on $\mathbb{C} - \{\sigma_j\}$ by Leibniz's rule. Our next goal will be to show that we can approximate every F_j by a rational function with poles only in $\{\sigma_j\}$. Fortunately, we have the following result.

<u>Lemma 2:</u> Suppose $K \subseteq \mathbb{C}$ is compact, $\gamma : [a,b] \to \mathbb{C}$ is a piecewise C^1 path (which guarentees it has finite length), and $\{\gamma\} \cap K = \emptyset$. If f is a continuous function on $\{\gamma\}$ and $\varepsilon > 0$ then there is a rational function $R(z)$ with poles only in $\{\gamma\}$ such that:

$$\left|\int_\gamma \frac{f(w)}{w-z}\mathrm{d}w - R(z)\right| < \varepsilon \text{ for all } z \in K$$

Proof:
Define $\varphi : \{\gamma\} \times K \to \mathbb{C}$ by $\varphi(w,z) = \frac{f(w)}{w-z}$. Note that φ is well-defined because $\{\gamma\} \cap K = \emptyset$ means that we will never have that $w - z = 0$. Also note that φ is continuous. And since $\{\gamma\} \times K$ is compact metric space (where we define the product metric $d((w_1,z_1),(w_2,z_2)) = \max(|w_1 - w_2|, |z_1 - z_2|)$), we can conclude that φ is uniformly continuous. Hence:

$$\forall \varepsilon > 0,\ \ \exists \delta > 0\ s.t.\ \forall (w_1,z_1),(w_2,z_2) \in \{\gamma\} \times K \text{ we have that}$$
$$|w_1 - w_2| < \delta \text{ and } |z_1 - z_2| < \delta \implies |\varphi(w_1,z_1) - \varphi(w_2,z_2)| < \varepsilon.$$

In particular, this means for any fixed $\varepsilon > 0$ we can find $\delta > 0$ such that for all $z \in K$ we have that $|w_1 - w_2| < \delta \implies |\varphi(w_1,z) - \varphi(w_2,z)| < \varepsilon$.

But note that as γ has finite length, we can subdivide γ into subpaths $\gamma_1, \ldots, \gamma_\ell$ of length $< \delta$. For each $1 \le s \le \ell$ pick $p_s \in \{\gamma_s\}$ and set $c_s = f(p_s)\int_{\gamma_s}\mathrm{d}w$. Finally, define $R(z) = \sum_{s=1}^\ell \frac{c_s}{p_s - z}$.

Clearly, R is a rational function with roots only in $\{\gamma\}$. (In particular R doesn't have a root at ∞.)

Also note for all $z \in K$ that:

$$\begin{aligned}\left|\int_\gamma \frac{f(w)}{w-z}\mathrm{d}w - R(z)\right| &= \left|\int_\gamma \frac{f(w)}{w-z}\mathrm{d}w - \sum_{s=1}^{\ell} \frac{f(p_s)}{p_s-z}\int_{\gamma_s}\mathrm{d}w\right| \\ &= \left|\sum_{s=1}^{\ell}\int_{\gamma_s}\left(\frac{f(w)}{w-z} - \frac{f(p_s)}{p_s-z}\right)\mathrm{d}w\right| \\ &\leq \sum_{s=1}^{\ell}\int_{\gamma_s}|\varphi(w,z)-\varphi(p_s,z)|\mathrm{d}|w| < \sum_{s=1}^{\ell}\varepsilon\cdot\mathrm{length}(\gamma_s) \\ &= \varepsilon\cdot\mathrm{length}(\gamma).\end{aligned}$$

By applying the prior lemma, for all $1 \leq j \leq n$ we can find a rational function R_j with poles only in $\{\sigma_j\}$ and satisfying that $|F_j - R_j| < \varepsilon/2n$ on K. Then because $f = \sum_{j=1}^n F_j$ we can conclude that $R = \sum_{j=1}^n R_j$ is a rational function with poles only in $\bigcup_{j=1}^n \{\sigma_j\}$ and satisfying that $|F - R| < \varepsilon/2$ on K.

Finally, we shall "push" the poles of R from $\bigcup_{j=1}^n \{\sigma_j\}$ to $\widehat{S}$.

Write $\mathbb{C}_\infty - K = \bigcup_\ell H_\ell$ where each H_ℓ is a connected component of $\mathbb{C}_\infty - K$.

<u>Lemma 3:</u> Suppose $H \subseteq \mathbb{C}$ is connected, $K \subseteq \mathbb{C}$ is compact, and $K \cap H = \emptyset$.

1. For all $a, b \in H$ we have that $\frac{1}{z-a}$ can be uniformly approximated on K by polynomials in $\frac{1}{z-b}$;
2. If H is unbounded then $\frac{1}{z-a}$ can be approximated uniformly on K by polynomials in z.

(Proof of 1.):
Fix b and consider the set:
$W := \{c \in H : \frac{1}{z-c}$ can be approximated uniformly on K by polynomials in $\frac{1}{z-b}\}$.

Note $W \neq \emptyset$ because $b \in W$. Also, we claim for all $c \in W$ that if $\delta := \inf\{|c-k| : k \in K\}$ (and as a reminder $\delta > 0$ since K is closed and $c \notin K$), then $\Delta(c,\delta) \subseteq W$.

Why?
Suppose $s \in \Delta(c,\delta)$ and consider the Laurent expansion of $\frac{1}{z-s}$ at c:

$$\frac{1}{z-s} = \frac{1}{z-c}\cdot\frac{1}{1-\frac{s-c}{z-c}} = \frac{1}{z-c}\sum_{k\geq 0}\left(\frac{s-c}{z-c}\right)^k = \sum_{k\geq 0}\frac{(s-c)^k}{(z-c)^{k+1}} \text{ if } |z-c| > |s-c|$$

Note that as $\Delta(c,\delta) \subseteq H$ and $K \cap H = \emptyset$, we can conclude that $|z-c| \geq \delta > |s-c|$ for all $z \in K$. Furthermore, as $\frac{|s-c|}{\delta} < 1$ we know that $\sum_{k\geq 0}\frac{|s-c|^k}{\delta^{k+1}} < \infty$. So by the Weierstrass M-test we can conclude that the above Laurent expansion converges uniformly on K.

Pick a Laurent polynomial T in $\frac{1}{z-c}$ from the expansion above so that $|\frac{1}{z-s} - T| < \frac{\varepsilon}{2}$ on K. Since $c \in W$, we know that $\frac{1}{z-c}$ can be approximated by polynomials in $\frac{1}{z-b}$. The same is then true about the polynomial T in $\frac{1}{z-c}$ on the compact set K.

This is because we can prove (if K is a compact set) that for any $f, g \in C(K)$, and sequences $\{f_n\}_{n\in\mathbb{N}}$, $\{g_n\}_{n\in\mathbb{N}}$ converging uniformly to f and g respectively that $f_n g_n \xrightarrow{u.} fg$. Also, since $f_n + g_n \xrightarrow{u.} f + g$ regardless of where K is compact or not, we can thus prove by induction our desired claim about being able to uniformly approximate T.

Let P be a polynomial in $\frac{1}{z-b}$ so that $|T - P| < \frac{\varepsilon}{2}$ on K. Then $|\frac{1}{z-s} - P| < \varepsilon$ on K, thus proving that $s \in W$.

It immediately follows that W and $H - W$ are open subset subsets of H. Since W is nonempty and H is connected, the only way this is possible is if $W = H$.

(Proof of 2.):
Pick $r > 0$ large enough so that $K \subseteq \Delta(0, r)$. Then by part 1 we can uniformly approximate $\frac{1}{z-a}$ on K by some polynomial T in $\frac{1}{z-c}$ where $|c| > r$. Then next we can consider Taylor polynomials of T near $z = 0$ in order to uniformly approximate T on the disk $\overline{\Delta}(0, r)$ containing K.

Now for all σ_j there exists a component H_ℓ with $\{\sigma_j\} \subseteq H_\ell$. After all, $\{\sigma_j\}$ is a connected subset of $\mathbb{C}_\infty - K$. In turn, we can apply the prior lemma to push the poles of R in σ_j to some specified point in $\widehat{S}$. By doing this for all j, we complete the proof. ■

Using entirely topological arguments, we can also formulate a version of Runge's theorem for open sets.

<u>Theorem:</u> Suppose $U \subseteq \mathbb{C}$ is an open set and $\widehat{S} \subseteq \mathbb{C}_\infty - U$ is a set containing at least one point from each connected component of $\mathbb{C}_\infty - U$. If $f \in O(U)$ then there exists a sequence of rational functions $\{R_n\}_{n\in\mathbb{N}}$ with poles only in $\widehat{S}$ such that $R_n \xrightarrow{\ell.u.} f$ on U.

Proof:
We start by proving the following lemma. The only nontrivial part of the lemma (which makes me unable to merely cite a result from math 240b) is proving part 4.

<u>Lemma:</u> For any $U \subseteq \mathbb{C}$ we can find a sequence of compact sets $K_n \subseteq U$ with:

1. $U = \bigcup_{n\geq 1} K_n$,
2. $K_n \subseteq K_{n+1}^\circ$,
3. for all compact $K \subseteq U$ there exists n with $K \subseteq K_n$,
4. *each* connected component of $\mathbb{C}_\infty - K_n$ contains a component of $\mathbb{C}_\infty - U$.

Proof:
If $U = \mathbb{C}$ then this lemma is easy. Just set $K_n = \overline{\Delta}(0, n)$ for all n. Hence, from now on we can assume without loss of generality that $U \neq \mathbb{C}$.

Let $K_n = \{z \in \mathbb{C} : |z| \leq n \text{ and } \inf\{|z - w| : w \in \mathbb{C} - U\} \geq \frac{1}{n}\}$ for all n.

- Note that if $z \in K_n$ then because $\inf\{|z - w| : w \in \mathbb{C} - U\} \geq \frac{1}{n} > 0$, we know that $z \notin \mathbb{C} - U$. This proves that $K_n \subseteq U$ for each n.

- Also note that if $z \in U$, then because $\inf\{|z-w| : w \in \mathbb{C} - U\} > 0$ since $\mathbb{C} - U$ is a disjoint closed set, we can find n with $\frac{1}{n}$ less than that infimum. Also by increasing n we can additionally get that $|z| \leq \frac{1}{n}$. Thus $z \in K_n$ and we've proven that $U = \bigcup_{n \geq 1} K_n$.

- Each K_n is clearly bounded. Also note that each K_n is closed since:
$$\mathbb{C} - K_n = \{z : |z| > n\} \cup \bigcup_{b \in \mathbb{C} - U} \Delta(b, \tfrac{1}{n})$$
is an arbitrary union of open sets. This proves each K_n is compact.

- Suppose $z \in K_n$ and let $r < \frac{1}{n} - \frac{1}{n+1}$. Then for all $w \in \Delta(z, r)$, we have that:
  - $|w| \leq |z| + |w - z| \leq n + r < n + 1$.
  - For all $a \in \mathbb{C} - U$ we have that $|z - a| \leq |w - z| + |w - a|$ implies that:
  $$|w - a| \geq |z - a| - |w - z| \geq \tfrac{1}{n} - r$$

  It follows that $\inf\{|w - a| : a \in \mathbb{C} - U\} \geq \frac{1}{n} - r > \frac{1}{n+1}$.

  As a result, $\Delta(z, r) \subseteq K_{n+1}$, thus proving that $z \in K_{n+1}^{\circ}$. THis proves that $K_n \subseteq K_{n+1}^{\circ}$ for all n.

- Note that if $K \subseteq U$ is any compact set, then the K_n° form an open cover of K. (After all, by our prior bullet point we know that:
$$U = \bigcup_{n \geq 1} K_n \subseteq \bigcup_{n \geq 2} K_n^{\circ} \subseteq \bigcup_{n \geq 2} K_n \subseteq U).$$

  It follows by the compactness of K that only finitely many of them are needed to cover K. Yet also as $\{K_n^{\circ}\}$ is an increasing sequence of sets, this in turn means that there is some n with $K \subseteq K_n^{\circ} \subseteq K_n$.

- Let $A = \mathbb{C}_{\infty} - K_n$ and $B = \mathbb{C}_{\infty} - U$. Then note that $A \supseteq B$. We claim that every connected component of A contains a connected component of B.

  Why?

  Let A_c be a component of A. Then firstly note that $\infty \in A \cap B$. As a result, if A_c is the component of A containing ∞, we can let B_c be the component of B also containing ∞. Then as $A_c \cap B_c \neq \emptyset$ and $B_c \subseteq B \subseteq A$, we can conclude that $B_c \subseteq A_c$.

  Meanwhile, suppose $\infty \notin A_c$ and recall from when we were showing K_n to be closed that:
$$A = (\{z : |z| > n\} \cup \{\infty\}) \cup \bigcup_{\substack{b \in B \\ b \neq \infty}} \Delta(b, \tfrac{1}{n})$$

  Now A_c can't be disjoint from every set $\Delta(b, \frac{1}{n})$ for all $b \in B - \{\infty\}$. After all, otherwise we'd have to have that A_c is contained in the connected set $(\{z : |z| > n\} \cup \{\infty\})$ which contains ∞. Yet this contradicts that $\infty \notin A_c$.

  As a result, there exists $b \in B - \{\infty\}$ with $A_c \cap \Delta(b, \frac{1}{n}) \neq \emptyset$. And since $\Delta(b, \frac{1}{n}) \subseteq A$ and is a connected set, this implies that $\Delta(b, \frac{1}{n}) \subseteq A_c$.

To finish off, let B_c be the component of B containing b. Then since $b \in B_c \cap A_c$ and $B_c \subseteq B \subseteq A$, we can see that $B_c \subseteq A_c$.

Using the prior topological lemma plus Runge's theorem for compact sets, hopefully it's now clear how to prove our desired theorem.

Suppose $f \in O(U)$ and let $\{K_n\}_{n\in\mathbb{N}}$ be an increasing sequence of compact sets in U constructed via the prior lemma. By applying Runge's theorem for compact sets to f and K_n, we can get a rational function R_n with poles only in $\widehat{S}$ and which satisfies that $|f - R_n| < \frac{1}{n}$ on K_n. It is easy to then check that $R_n \xrightarrow{\ell.u.} f$ on U. ■

Here is the final theorem of Math 220b:

<u>Theorem:</u> Suppose $G \subseteq \mathbb{C}$ is open and connected. Then the following are equivalent.

(a) G is simply connected;

(b) for all piecewise C^1 loops we have that $\gamma \approx_G 0$ (recall *page 426* for what this means);

(c) $\mathbb{C}_\infty - G$ is connected,

(d) any function $f \in O(G)$ can be approximated locally uniformly in G by polynomials;

(e) for all piecewise C^1 loops γ and $f \in O(G)$, we have that $\int_\gamma f \mathrm{d}z = 0$;

(f) any $f \in O(G)$ admits a primitive (i.e. some $F \in O(G)$ with $F' = f$);

(g) for all $f \in O(G)$ that are nowhere zero, we have that $f = e^g$ for some $g \in O(G)$;

(h) for all $f \in O(G)$ that are nowhere zero, we have that $f = h^2$ for some $h \in O(U)$;

(i) G is homeomorphic to $\Delta(0,1)$.

Proof:
We've already shown most of the implications here.

$(a \implies b)$
This is just a consequence of the fact that $\gamma \sim_G 0 \implies \gamma \approx_G 0$ (as shown on *page 415*).

$(c \implies d)$
This is a consequence of Runge's theorem for open sets that we just proved.

$(d \implies e)$
Let $\{p_n\}_{n\in\mathbb{N}}$ be a sequence of polynomials converging to f locally uniformly. Then $p_n \xrightarrow{u.} f$ on $\{\gamma\}$. In turn, $\int_\gamma f \mathrm{d}z = \lim_{n\to\infty} \int_\gamma p_n \mathrm{d}z = \lim_{n\to\infty} 0 = 0$.
(Note that $\int_\gamma p_n \mathrm{d}z = 0$ because p_n admits a primitive on G...)

$(e \implies f)$
Fix $a \in G$ and define $F(z) = \int_\gamma f \mathrm{d}w$ where γ is any piecewise C^1 path from a to z. By part (e) we can see that $F(z)$ is independent of the particular γ chosen. Then by identical reasoning to the proof of the theorem on *pages 415-416*, we can prove that $F' = f$.

$(f \implies g)$
The proof of this is identical to the proof of the theorem on *page 416*.

$(g \implies h)$
Write $f = e^g$ and set $h := e^{g/2}$.

$(h \implies i)$
If $G \neq \mathbb{C}$ then we have by the Riemann mapping theorem that G and $\Delta(0,1)$ are biholomorphic and hence homeomorphic. Meanwhile, if $G = \mathbb{C}$ then we can consider the homeomorphism $z \mapsto \frac{z}{\sqrt{1+|z|^2}}$ from $\mathbb{C}$ to $\Delta(0,1)$

To justify why this is a homeomorphism, first note that if $w = \frac{z}{\sqrt{1+|z|^2}}$ and $w, z \neq 0$ then $\operatorname{sgn}(w) = \operatorname{sgn}(z)$. Also note that:

$$|w| = \frac{|z|}{\sqrt{1+|z|^2}} \implies |w|^2 = \frac{|z|^2}{1+|z|^2} \implies |z|^2 = \frac{|w|^2}{1-|w|^2} \implies |z| = \frac{|w|}{\sqrt{1-|w|^2}}$$

Thus, we can see that the inverse of our proposed homeomorphism is $w \mapsto \frac{w}{\sqrt{1-|w|^2}}$. Now this and the function $z \mapsto \frac{z}{\sqrt{1+|z|^2}}$ are continuous on the domains we want. So, they are homeomorphisms.

$(i \implies a)$
Since Δ is simply connected, if G is homeomorphic to Δ then G is also simply connected.

The only implication which I haven't really proven in the past is that (b) implies (c).

Suppose for the sake of contradiction that $\mathbb{C}_\infty - G = A \cup B$ where A, B are nonempty disjoint closed sets in $\mathbb{C}_\infty$.

We can assume without loss of generality that A and B are closed in $\mathbb{C}_\infty$ as opposed to merely being closed in $\mathbb{C}_\infty - G$ because $\mathbb{C}_\infty - G$ is closed.

Also, without loss of generality assume that $\infty \in B$. Then as $\mathbb{C}_\infty - A$ is an open set in $\mathbb{C}_\infty$ containing ∞, we can guarentee from the definition of the one-point compactification of $\mathbb{C}$ that A is a compact subset of $\mathbb{C}$. Also note that $V := G \cup A = \mathbb{C}_\infty - B$ is an open subset of $\mathbb{C}_\infty$ not containing ∞. Thus V is an open subset of $\mathbb{C}$ containing A.

By the first lemma in the proof of Runge's theorem for compact sets, we can find oriented line segments $\sigma_1, \ldots, \sigma_n$ in $V - A = G$ with $f(a) = \frac{1}{2\pi i} \sum_{j=1}^n \int_{\sigma_j} \frac{f(z)}{z-a} \mathrm{d}z$ for all $a \in A$ and $f \in O(V)$. By cleverly joining the σ_j, we can then get closed polygonal curves $\pi_1, \ldots, \pi_m \in V - A = G$ with $f(a) = \frac{1}{2\pi i} \sum_{j=1}^m \int_{\pi_j} \frac{f(z)}{z-a} \mathrm{d}z$ for all $a \in A$ and $f \in O(V)$.

Frick I ought to formally justify this. I do not have time to do this right now but I will come back to this on *page* ____. It will be worth formally justifying this though because it will involve some wall hugging reasoning which I have no practice with.

Fix f to be the constant function 1. Then $1 = \sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\sigma_j} \frac{\mathrm{d}z}{z-a} = \sum_{j=1}^{m} n(\pi_j; a)$. But by (b) we know the last sum equals 0 since each π_j is a piecewise C^1 loop in G and $a \in A \subseteq \mathbb{C} - G$. This is a contradiction. ■

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Set 8 Problem 8: suppose $\alpha, \beta \in \mathbb{C}$ are algebraic over $\mathbb{Q}$. Let $f := m_{\alpha;\mathbb{Q}}$ and $g := m_{\beta;\mathbb{Q}}$. Prove that f is irreducible in $(\mathbb{Q}[\beta])[x]$ if and only if g is irreducible in $(\mathbb{Q}[\alpha])[x]$.

To start off, we show that f is irreducible in $(\mathbb{Q}[\beta])[x]$ iff $[\mathbb{Q}[\alpha, \beta] : \mathbb{Q}] = \deg(f)\deg(g)$.

Note that if f is irreducible in $(\mathbb{Q}[\beta])[x]$, then as $f(\alpha) = 0$ and f is monic we have that $f = m_{\alpha;\mathbb{Q}[\beta]}$. Hence by the tower formula we can conclude that:

$$[\mathbb{Q}[\alpha, \beta] : \mathbb{Q}] = [\mathbb{Q}[\alpha, \beta], \mathbb{Q}[\beta]] \cdot [\mathbb{Q}[\beta] : \mathbb{Q}] = \deg(f)\deg(g).$$

Conversely, suppose $[\mathbb{Q}[\alpha, \beta] : \mathbb{Q}] = \deg(f)\deg(g)$. Then by the tower formula we can guarentee that $\deg(m_{\alpha;\mathbb{Q}[\beta]}) = \frac{[\mathbb{Q}[\alpha,\beta]:\mathbb{Q}]}{\deg(g)} = \deg(f)$. Since $m_{\alpha;\mathbb{Q}[\beta]}$ divides f (as both evaluate to 0 at α) and both polynomials are monic, we can conclude that $f = m_{\alpha;\mathbb{Q}[\beta]}$. And this proves that f is irreducible in $\mathbb{Q}[\beta]$.

By symmetric reasoning, we have that g is irreducible in $(\mathbb{Q}[\alpha])[x]$ iff $[\mathbb{Q}[\alpha, \beta] : \mathbb{Q}] = \deg(f)\deg(g)$. And this let's us conclude what we wanted. ■

Set 8 Problem 7: Prove that $\mathbb{Q}[\sqrt{2}]$ and $\mathbb{Q}[\sqrt{3}]$ are not isomorphic (as fields).

Suppose for the sake of contradiction that $\theta : \mathbb{Q}[\sqrt{2}] \to \mathbb{Q}[\sqrt{3}]$ is a field isomorphism, then note that since $\theta(1) = 1$ and θ preserves sums and differences, we can conclude that $\theta(n) = n$ for all $n \in \mathbb{Z}$. In turn, since θ preserves division we must have that $\theta(q) = q$ for all $q \in \mathbb{Q}$.

Next, consider that $\theta(\sqrt{2})^2 = \theta(2) = 2$. Yet simultaneously since the minimal polynomial of $\sqrt{3}$ over $\mathbb{Q}$ is $x^2 - 3$, we know that $\{1, \sqrt{3}\}$ is a $\mathbb{Q}$-basis of $\mathbb{Q}[\sqrt{3}]$ and in turn that $\theta(\sqrt{2}) = p + q\sqrt{3}$ for some $p, q \in \mathbb{Q}$. So, we can conclude that:

$$p^2 + 2pq\sqrt{3} + 3q^2 = 2$$

Equivalently, this means that $(p^2 + 3q^2 - 2) + (2pq)\sqrt{3} = 0$. And since 1 and $\sqrt{3}$ are $\mathbb{Q}$-linearly independent, we know $p^2 + 3q^2 - 2 = 0$ and $2pq = 0$. But now this implies that either $p = 0$ or $q = 0$. We can check as follows that both of these possibilities lead to a contradiction.

- If $q = 0$ then we must have that $p^2 = 2$. Yet this is impossible as no $p \in \mathbb{Q}$ satisfies that $p^2 = 2$.
- If $p = 0$ then we have that $q^2 = \frac{2}{3}$ Yet once again no $q \in \mathbb{Q}$ satisfies this.

Thus we conclude no such θ can exist. ■

Set 7 Problem 1: Prove that $\langle 2, \sqrt{-10} \rangle$ is a projective $\mathbb{Z}[\sqrt{-10}]$-module.

Firstly, consider that the minimal polynomial of $\sqrt{-10}$ in $\mathbb{Q}[x]$ is $m_{\sqrt{-10};\mathbb{Q}}(x) = x^2 + 10$. (For a reminder of how this is defined)